

Lakshya Aryan

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EDUCATION

Indian Institute Of Technology Varanasi (BHU)

Bachelor of Technology

Varanasi, UP

Nov 2022 – Present

EXPERIENCE

Quantitative Researcher Internship

Jun 2025 – Sep 2025

AR FINANCIAL SECURITIES

Remote

- Enhanced option pricing accuracy using Variance Gamma model beyond Black-Scholes improving pricing stability
- Processed data for 2,500 stocks across key sectors and built a dashboard with visualizations for actionable insights.
- Designed a KAMA-based portfolio construction strategy across multiple stocks to enhance and improve metrics.

Quantitative Developer Internship

May 2025 – June 2025

SPRINGPAD WEALTH

Remote

- Built ML-driven models (Random Forest, XGBoost) to forecast Nifty & BankNifty trends using technical Indicator
- Designed and backtested a Z-score mean-reversion pairs trading strategy on Gold vs. Silver with 5 years of data.
- Delivered 54.06% returns with Sharpe of 0.67 while maintaining strict risk control at -16.86% max drawdown.

PROJECTS

Price Action Trading Strategy | *Inter-IIT Tech Meet*

- Built a price-action signal system achieving 99.8 retracement accuracy across over 60 Indian equities overall result.
- Applied hypothesis testing to map price-time and price-volume domains, effectively reducing false signals further.
- Backtested the strategy on liquid Indian stocks and optimized entry and exit rules for consistent profits further.

Trade Matching Engine | *C++, STL, Multithreading*

- Engineered a high-performance order matching system for crypto trading with FIFO and price-priority algorithm.
- Designed scalable architecture using Red-Black trees, hash-maps for price levels and linked lists for order queues.
- Achieved $O(1)$ order cancel/execute and $O(\log M)$ through optimized data structure choices for high performance.

Volatility Index Estimation Calculator | *Python, NumPy, Pandas*

- Implemented end-to-end VIX computation using CBOE, integrating options data, yield curves, and expiry logic.
- Engineered automated strike selection, OTM filtering, and variance-swap sigma modeling for reliable estimations.
- Optimized data processing for forward pricing, sigma aggregation, and final VIX estimation across timeframes.

POSITIONS OF RESPONSIBILITY

Operations Manager, Technex'24

- Managed operational logistics, on-ground coordination, and overall event workflow to support Technex efficiently.
- Led cross-team communication to ensure smooth execution of Technex, one of largest and most active fest.

Marketing Executive, Kashiyatra'23

- Coordinated sponsorship outreach and partnership activities while driving key marketing initiatives across teams
- Secured key sponsors and supported outreach efforts, enhancing visibility and expanding the overall event scale.

HONOURS & ACHIEVEMENTS

- *Star Quant of the Month* – for exceptional performance during internship.
- Certificate of Excellence in Machine Learning & Summer Analytics Program, *IIT Guwahati*.
- National Runner-up, Quantalytics *IIT Jodhpur* – built and backtested a quant strategy.
- Achieved *1744 max LeetCode rating*, demonstrating strong problem-solving skills.
- Participated in *HCL Tech Hackathon (GenAI Track)* and selected for PPO as Senior AI Scientist.